Pediatric Bradycardia (<13 years)

Bradycardia Identified

Consider ALS Back-Up

Cardiopulmonary Compromise?
- Acutely Altered Mental Status
- Signs of Shock
- Hypotension

YES

Maintain Patent Airway
(OPA/NPA/LT airway if required)

EMT

Assist Breathing
with positive pressure ventilation and oxygen as necessary

Cardiac Monitor
To identify rhythm, monitor pulse rate, BP and Sp02

NO

Support Airway, Breathing and Circulation
- Consider oxygen
- 12-lead ECG
- Identify and treat underlying causes
- Focus on adequate oxygenation and ventilation

Heart Rate < 60bpm
Despite oxygenation and ventilation?

YES

Start CPR

Continue CPR if HR<60

Pediatric Cardiac Arrest CPG

IV/IO Access

EMT

Adrenaline IV/IO
0.01mg/kg (every 3-5minutes)

EMT

Consider Atropine IV/IO
0.02mg/kg
if increased vagal tone or primary AV block

EMT-A

NO

Support Airway, Breathing and Circulation
- Consider oxygen
- 12-lead ECG
- Identify and treat underlying causes
- Focus on adequate oxygenation and ventilation

Patient Care Goals

- Recognize and promptly treat symptomatic bradycardia to prevent progression to cardiac arrest.
- Identify and address reversible causes.
- Support oxygenation, ventilation, and perfusion.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Age: Infants and children ≤13 years.
- Heart rate below normal for age, with associated signs of poor perfusion:
    - Altered mental status.
    - Pallor, cyanosis, weak pulses, prolonged capillary refill.
    - Hypotension or respiratory failure.
- Bradycardia due to reversible causes:
    - Hypoxia, vagal stimulation, hypothermia, toxins, heart block.
- Unstable bradycardia: Not tolerating low HR, progressing to shock or cardiac arrest.

Exclusion Criteria

- Heart rate appropriate for age without signs of hypoperfusion.
- Known congenital heart disease with stable baseline bradycardia without deterioration.

Key Patient Assessment Points

1. **Initial impression**
    - Assess responsiveness, appearance, and distress using the Pediatric Assessment Triangle (PAT): tone, interactiveness, breathing effort, skin color.
    - Look for signs of instability: lethargy, cyanosis, poor perfusion.
2. **Airway & Breathing Assessment**
    - Ensure airway is patent and assess for respiratory distress or failure.
    - Look for hypoxia, apnea, or shallow respirations, the most common cause of bradycardia.

Normal Respiratory Rate Ranges[1]	
Age	**Breathing Rate** (breaths/min)
1-12 months	30-53
1-3yrs	22-37
3-5yrs	20-28
5-11yrs	18-25
>11yrs	12-20

3. **Circulation Assessment**
    - Palpate central and peripheral pulses for rate, rhythm, and quality.
    - Check capillary refill time (>3 seconds suggests poor perfusion).
    - Measure blood pressure if possible.
    - Look for skin signs: pallor, mottling, cyanosis.

Hypotension[1]	
Age	**Systolic BP**
<1 month	<60
1-12 months	<70

1-10yrs	<(70+(age yrs x 2))
>10yrs	<90

4. **Heart Rate Assessment**
   - Count HR accurately over a full minute.
   - Compare to age-appropriate normal ranges:

Normal Heart Rate Ranges[1]	
Age	**Pulse Rate (beats/min)**
<1 month	100-205
1-12 months	100-180
1-3yrs	100-140
3-5yrs	80-120
5-11yrs	75-120
>11yrs	60-100

   - A heart rate below these thresholds with poor perfusion is significant and warrants immediate intervention.
5. **ECG Monitoring**
   - Apply ECG to identify rhythm (e.g., sinus bradycardia, heart block).
6. **Assess for Reversible Causes**
   - Such as; hypoxia, hypovolemia, hypothermia, toxins/drugs, increased vagal tone, or elevated intracranial pressure.
7. **Patient History**
   - Assess for alternative pathology or events which may guide treatment, such as; recent choking, respiratory infection, toxin/medication ingestion, congenital heart disease, prior episodes.
8. **Level of consciousness**
   - Assess alertness, confusion, lethargy, or unresponsiveness, indicating cerebral hypoperfusion.
2. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
   - Red Priority:
     - Bradycardia with poor perfusion, altered mental status, hypotension, or impending cardiac arrest.
     - Requires immediate intervention.
   - Yellow Priority:
     - Bradycardia with stable perfusion, normal mental status, and no signs of shock.
     - Continue close monitoring and be prepared to escalate if deterioration occurs.

Patient Management & Interventions

1. **Request ALS back-up**
   - Request ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested for pediatric patients with bradycardia.
1. **Maintain a Calm, Systematic Approach**
   - Minimize stress and keep the child warm.
   - Allow caregiver to stay close if it helps calm the child.
2. **Support Oxygenation & Ventilation**
   - Ensure a patient airway.
   - Administer high-flow oxygen to maintain Spo2 >94%.
   - If respiratory effort is inadequate or apnea is present, start bag-valve-mask ventilation with 100% oxygen.

- Correct hypoxia promptly, most pediatric bradycardia is secondary to hypoxia.

3. **Start CPR if Indicated**
   - If HR <60 bpm with signs of poor perfusion despite oxygenation & ventilation, begin chest compressions immediately.
   - Use appropriate compression depth and ratio for age (15:2 if two rescuers, 30:2 if alone).
4. **Obtain IV/IO access**
   - Establish intravenous or intraosseous access as soon as possible.
5. **Administer Adrenaline (Epinephrine)**
   - If bradycardia persists despite adequate oxygenation & ventilation:
     - IV/IO: 0.01 mg/kg of 1:10,000 (0.1 mL/kg) every 3–5 minutes as needed.
     - If no IV/IO access, via ETT: 0.1 mg/kg of 1:1,000 diluted in 3–5 mL saline.
6. **Administer Atropine**
   - If vagal-mediated or suspected heart block:
     - Dose: Atropine 0.02 mg/kg IV/IO(minimum 0.1 mg; maximum single dose 0.5 mg).
     - May repeat once after 3–5 minutes.
     - Maximum total dose: 1 mg in children, 3 mg in adolescents.
7. **Advanced Airway**
   - Prepare for endotracheal intubation if respiratory failure persists or airway is not maintained.
8. **Consider Transcutaneous Pacing**
   - If refractory to oxygenation, ventilation, chest compressions, epinephrine, and atropine, and if equipment and physician or advance paramedic are available, consider transcutaneous pacing.
9. **Identify & Treat Reversible Causes**
   - Hypoxia → oxygen & ventilation.
   - Hypothermia → rewarming.
   - Toxins → antidotes if appropriate.
   - Vagal stimulation → remove stimulus.
10. **If Pulselessness Develops:**
    - Go to Pediatric Cardiac Arrest CPG.
11. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.
    - Specific Patient Reassessment Points:
      - Continuous ECG monitoring is required:
        - Reassess for profound bradycardia, cardiac arrest, or worsening perfusion.
        - Be prepared to resume or escalate resuscitation efforts if condition worsens.

Patient Safety Considerations

- Avoid excessive ventilation, maintain normocapnia.
- Aim to maintain normocapnia, with an end-tidal CO_2 ($EtCO_2$) between 35–45 mmHg (4.6–6 kPa).
- Minimize delays in initiating chest compressions when indicated.
- Use appropriate drug dosing based on weight, the use of Braslow scale
- Recognize bradycardia may indicate impending cardiac arrest in children.

Documentation

- Complete a comprehensive electronic Patient Care Record (ePCR), capturing the following elements to support continuity of care. Including:
  - Identification: Patient's age, sex, weight.
  - Mechanism/Medical complaint: Suspected cause of bradycardia.
  - Injuries/Information: Associated findings, vital signs.
  - Signs: Heart rate, blood pressure, SpO_2, mental status.
  - Treatment: Oxygen, medications, CPR, airway interventions.
  - Allergies: Known drug or food allergies.

- Medications: Medications administered prehospital.
- Background: Relevant medical history.
- Other: Time of onset, response to treatment, handover to hospital.

Key Performance Indicators

1. Start CPR for unconscious patients with HR <60bpm.
2. Administer oxygen for patients with SpO_2 <94%.

Clinical Insights & Additional Guidance

- Pediatric bradycardia is usually secondary to hypoxia, ensure oxygenation and ventilation first.
- Bradycardia may precede cardiac arrest , act promptly.
- Consult TeleEMS if in doubt, if the patient does not respond to initial measures, or if advanced interventions (e.g., pacing) may be needed.
- Distinguish between stable sinus bradycardia and symptomatic, unstable bradycardia requiring intervention.

References

1. American Heart Association, 2020. *AHA pediatric advanced life support (PALS) guidelines*. Available at: https://www.heart.org.
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5. National Association of EMS Officials (NAEMSO), 2022. *National model EMS clinical guidelines*. Available at: https://naemso.org.
6. Oxford University Press, 2007. *Oxford handbook of pre-hospital care*, 1st edition. Oxford: Oxford University Press.